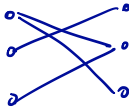


The Bipartite Matching Problem

$$G = (V, E), \quad V = V_1 \sqcup V_2$$

$$E \subseteq V_1 \times V_2$$



Thm: A graph G is bipartite iff any cycle has an even length.

• For any subset $S \subseteq E$, let $x_e^S \in [0, 1]^{|E|}$, $x_e^S = \begin{cases} 1 & e \in S \\ 0 & \text{o.w.} \end{cases}$

$$P_{\text{match}}(G) \triangleq \text{conv} \{ x^M : M \text{ is a matching} \} \subseteq \mathbb{R}^{|E|}$$

$M \subseteq E$ is a matching if no two edges are incident on the same vertex. $\forall v \in V$, $\sum_{e \in \delta(v)} x_e^M \leq 1$. $\leftarrow$

LP Relaxation

$$P_{\text{match}}(G) = \text{conv} \left(\begin{array}{ll} \sum_{e \in \delta(v)} x_e \leq 1 & \forall v \\ x_e \in [0, 1] & \forall e \end{array} \right)$$

$$Q(G) = \{x \in \mathbb{R}^{|E|} : 0 \leq x_e \leq 1 \\ \sum_{e \in \delta(v)} x_e \leq 1\}$$

$$Q(G) \supseteq P_{\text{match}}(G)$$

The LP Relaxation is Exact

Thm: If G is bipartite,

$$\text{then, } Q(G) = P_{\text{match}}(G).$$

$$Q(G) \geq P_{\text{match}}(G)$$

Proof — 2/5

To show $Q(G) \subseteq P_{\text{match}}(G)$,

take $\alpha \in Q(G)$

Let $F = \text{supp}(\alpha)$, i.e., $F \subseteq E$
 $e \in F \iff x_e > 0$.

Proof is by induction on $|F|$.

Base case: $|F| \leq 1$.

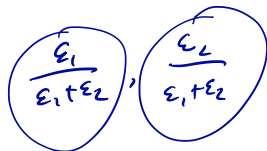
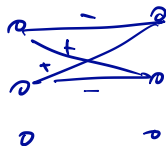
Proof — 3/5

Suppose F has a cycle,

G bipartite $\Rightarrow$ cycle must be even.

Let $d \in \{0, +1, -1\}^E$, $+1 \leq -1$

alternating along the cycle.



$$\text{Supp}(d) \subseteq F \subseteq \text{supp}(x)$$

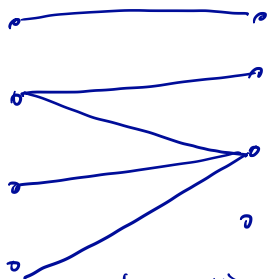
$$\sum_{e \in \delta(v)} (x \pm \epsilon d)_e = \sum_{e \in \delta(v)} x_e \pm \epsilon \sum_{e \in \delta(v)} d_e \leq 1$$

If ϵ small enough: $x \pm \epsilon d \geq 0$

Take ϵ_1, ϵ_2 as large as possible s.t. $x_1 = x + \epsilon_1 d \in Q(G)$
 $x_2 = x - \epsilon_2 d \in Q(G)$

Note: $|\text{supp}(x_i)| \leq |F| - 1$, $i=1,2 \Rightarrow x_1 = \sum \alpha_n x^n$
 $x_2 = \sum \beta_n x^n$

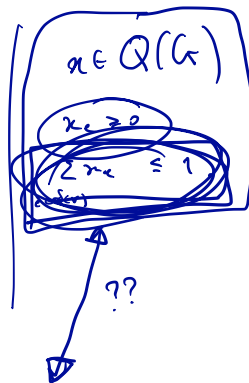
Proof — 4/5



F has no cycles
 $\Rightarrow F$ is a forest
 but not a matching.

So $\exists \tilde{P}$ a path

of maximal length, ≥ 2 .



Claim: $e \in \tilde{P}$,

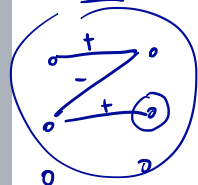
$$0 < x_e < 1$$

[Ex: why $x_e < 1$?]

d same as prev slide.

As before: For $\varepsilon > 0$ small enough, $x \pm \varepsilon d \in Q(G)$

Proceed as before.



Proof — 5/5

F is a matching.
"supplex".

Need to write: $x = \sum \lambda_i x^{M_i}$
 $\lambda_i \geq 0, \sum \lambda_i = 1.$

Order x : $x_1 \leq x_2 \leq \dots \leq x_{|F|}$

$$x = x_1 \cdot x^{F_1} + (x_2 - x_1) \cdot x^{F_2} + \dots$$

$$F_i = \{i, i+1, \dots\}, \quad F_i \subseteq F, \quad F \text{ a matching} \\ \Rightarrow F_i \text{ a matching}$$

$$\Rightarrow x \in Q(G) \Rightarrow x \in P(G)$$

$$\Rightarrow Q(G) = P(G) \quad \square$$

Consequence :

G bipartite :

$$\max: \sum x_e$$

$$\text{st: } x \in Q(G)$$

Solution gives
value of
max size
matching.